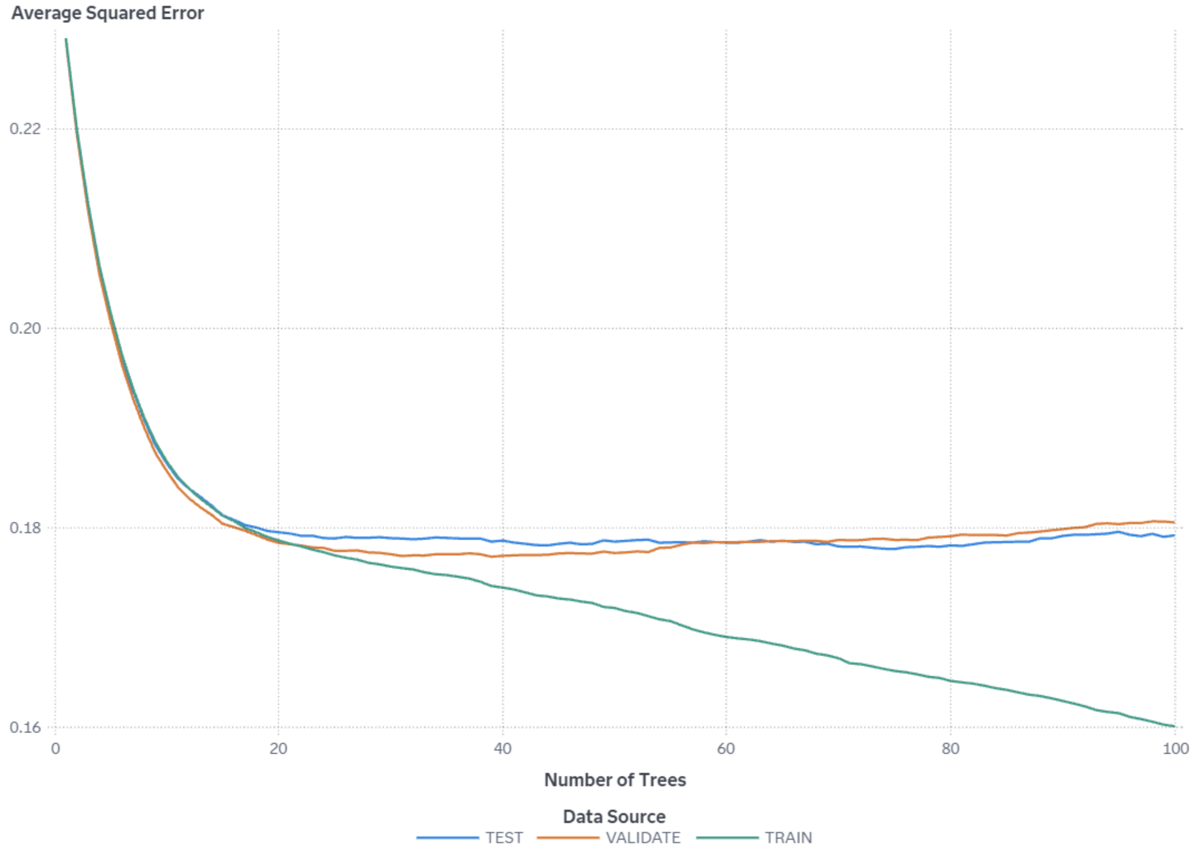
Delivery Delay Prediction in Logistics Systems

"Gradient Boosting" Results

by: ci240040@student.uthm.edu.my

Contents

Average Squared Error

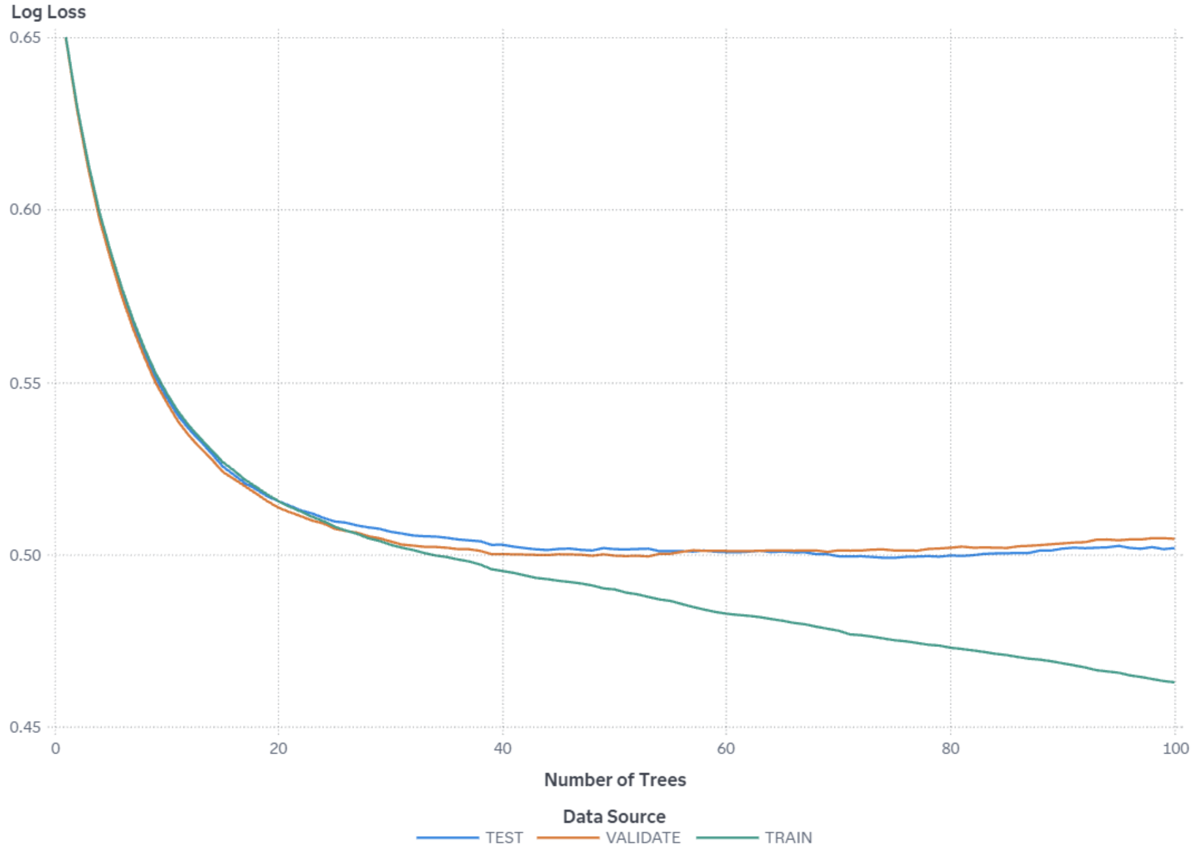


This plot shows how the average squared error changes as the number of trees in the gradient boosting model increases. If there are fewer trees than specified in the "Number of trees" property, then training stopped due to early stopping criteria. The training error decreases as the number of trees increases, but if the validation error shows an increase after a decrease, this could be a sign of overfitting. For this model, the minimum error for the VALIDATE partition is 0.177 and occurs for 39 trees, so early-stopping options might potentially be used or modified to prevent overfitting.

Misclassification Rate

This plot shows how the misclassification rate changes as the number of trees in the gradient boosting model increases. If there are fewer trees than specified in the "Number of trees" property, then training stopped due to early stopping criteria. The training error decreases as the number of trees increases, but if the validation error shows an increase after a decrease, this could be a sign of overfitting. For this model, the minimum error for the VALIDATE partition is 0.309 and occurs for 37 trees, so early-stopping options might potentially be used or modified to prevent overfitting.

Log Loss



This plot shows how the log loss changes as the number of trees in the gradient boosting model increases. If there are fewer trees than specified in the "Number of trees" property, then training stopped due to early stopping criteria. The training error decreases as the number of trees increases, but if the validation error shows an increase after a decrease, this could be a sign of overfitting. For this model, the minimum error for the VALIDATE partition is 0.5 and occurs for 53 trees, so early-stopping options might potentially be used or modified to prevent overfitting.

Variable Importance

Variable Label	Role	Variable Name	Training Importance	Importance Standard Deviation	Relative Importance
	INPUT	Discount_offered	24.7270	54.7061	1
	INPUT	Weight_in_gms	6.7128	7.5593	0.2715
	INPUT	Prior_purchases	5.3399	8.1657	0.2160
	INPUT	Cost_of_the_Product	4.3474	3.0244	0.1758
	INPUT	Customer_rating	2.5987	2.4439	0.1051
	INPUT	Customer_care_calls	2.4959	2.3762	0.1009
	INPUT	Warehouse_block	2.0633	2.0216	0.0834
	INPUT	Product_importance	0.8122	1.3192	0.0328
	INPUT	Mode_of_Shipment	0.6460	1.1762	0.0261
	INPUT	Gender	0.4156	0.9089	0.0168

Score Inputs

Name	Role	Variable Level	Type
Cost_of_the_Product	INPUT	INTERVAL	N
Customer_care_calls	INPUT	NOMINAL	N
Customer_rating	INPUT	NOMINAL	N
Discount_offered	INPUT	INTERVAL	N
Gender	INPUT	NOMINAL	C
Mode_of_Shipment	INPUT	NOMINAL	C
Prior_purchases	INPUT	NOMINAL	N
Product_importance	INPUT	NOMINAL	C
Warehouse_block	INPUT	NOMINAL	C
Weight_in_gms	INPUT	INTERVAL	N

Variable Type	Variable Label	Variable Format	Variable Length
double			8
double			8
double			8
double			8
varchar			1
varchar			6
double			8
varchar			6
varchar			1
double			8

Score Outputs

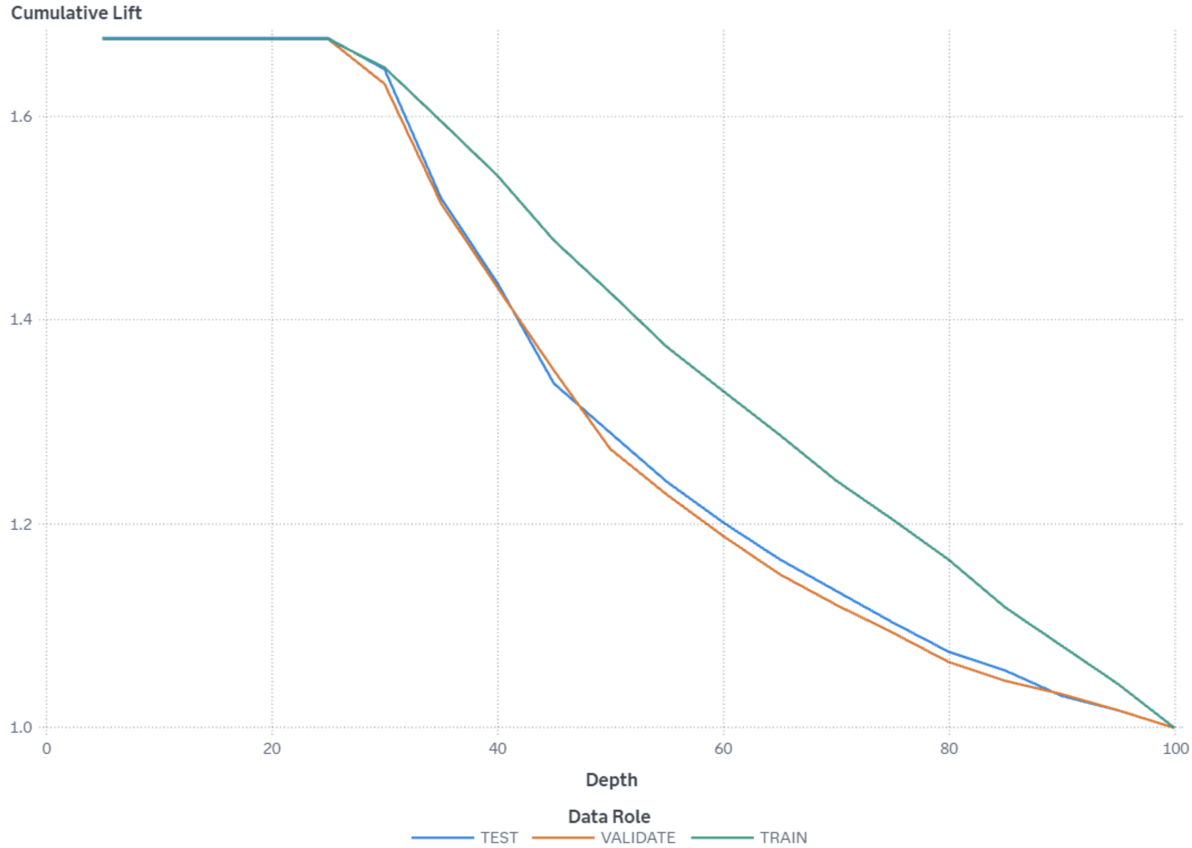
Name	Role	Type	Variable Type
EM_CLASSIFICATION	CLASSIFICATION	C	char
EM_EVENTPROBABILITY	PREDICT	N	double
EM_PROBABILITY	PREDICT	N	double
I_Reached_on_Time_Y_N	CLASSIFICATION	C	char
P_Reached_on_Time_Y_N0	PREDICT	N	double
P_Reached_on_Time_Y_N1	PREDICT	N	double
WARN	ASSESS	C	char

Variable Label	Variable Format	Variable Length	Creator
Predicted for Reached.on.Time_Y.N		12	gradboost
Probability for Reached.on.Time_Y.N=1		8	gradboost
Probability of Classification		8	gradboost
Into: Reached.on.Time_Y.N		12	gradboost
Predicted: Reached.on.Time_Y.N=0		8	gradboost
Predicted: Reached.on.Time_Y.N=1		8	gradboost
Warnings		4	gradboost

Function	Creator GUID
CLASSIFICATION	ff38b098-862b-4d8

Function	Creator GUID
	a- b6d5-848f598b967 3
PREDICT	ff38b098-862b-4d8 a- b6d5-848f598b967 3
PREDICT	ff38b098-862b-4d8 a- b6d5-848f598b967 3
CLASSIFICATION	ff38b098-862b-4d8 a- b6d5-848f598b967 3
PREDICT	ff38b098-862b-4d8 a- b6d5-848f598b967 3
PREDICT	ff38b098-862b-4d8 a- b6d5-848f598b967 3
ASSESS	ff38b098-862b-4d8 a- b6d5-848f598b967 3

Cumulative Lift



The VALIDATE partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

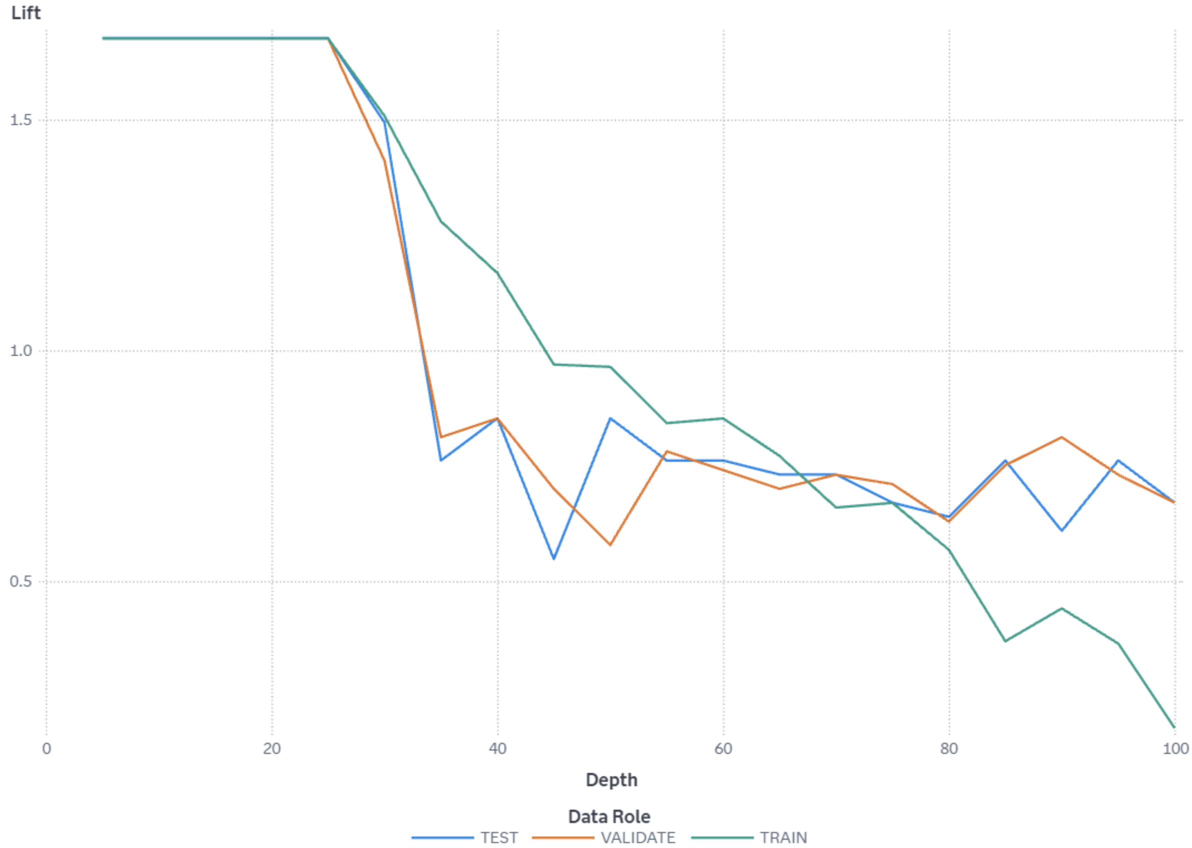
The TRAIN partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Cumulative Lift of 1.68 in the 10% quantile (depth of 10) meaning there are 1.68 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target

Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.

Lift



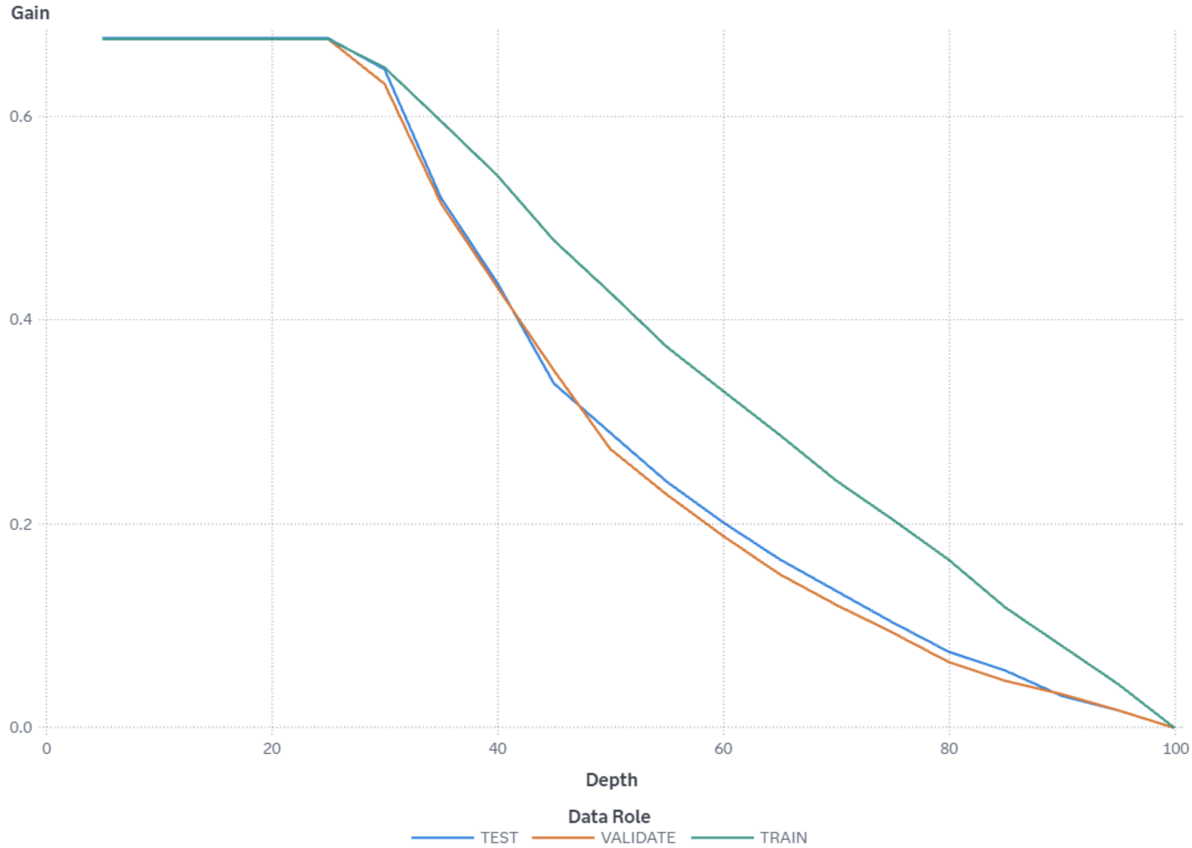
The VALIDATE partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Lift of 1.68 in the 5% quantile (depth of 5) meaning there are 1.68 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Lift is the ratio of the number of events in that quantile to the number of events that would be there at random, or equivalently, the ratio of the response percentage to the baseline response percentage. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Thus, Lift measures how much more likely it is to observe an event in each quantile than by selecting observations at random.

Gain



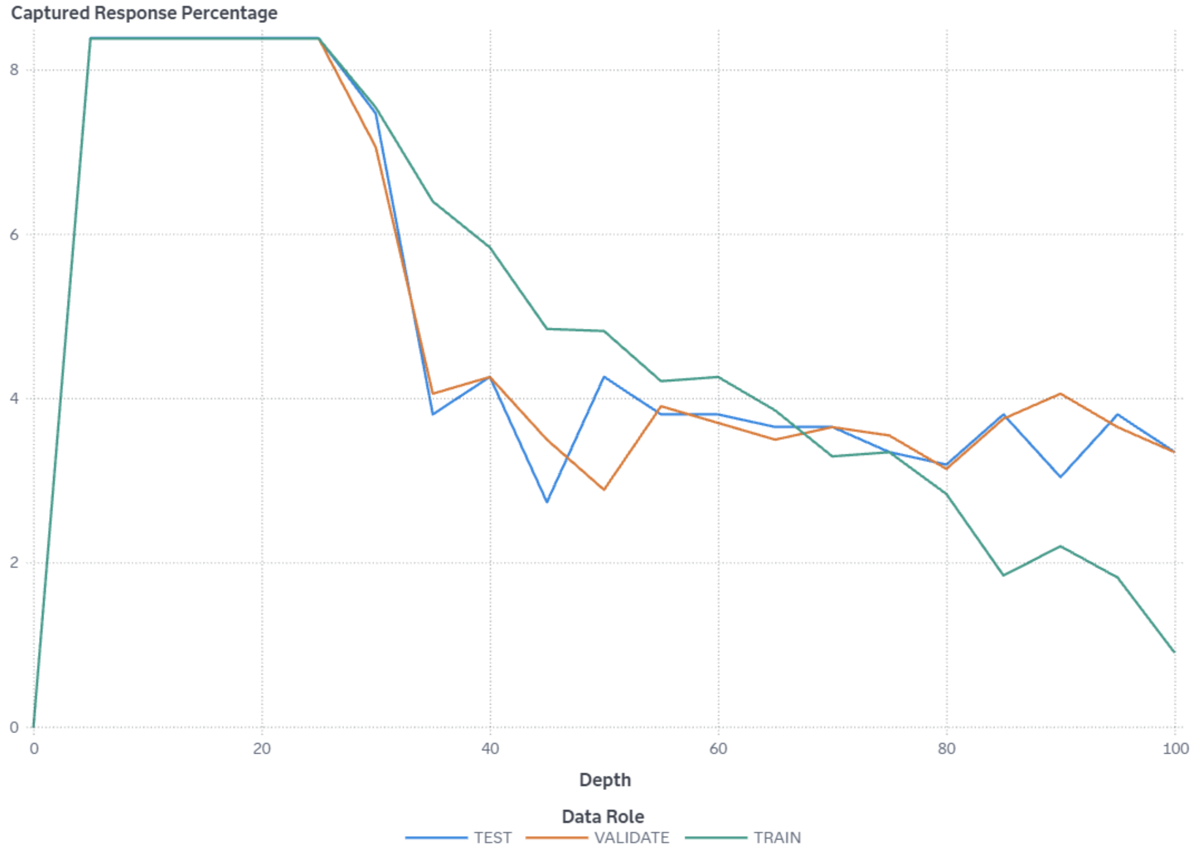
The VALIDATE partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

The TRAIN partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

The TEST partition has a Gain of 0.7 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

Gain is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Gain is a cumulative measure for the quantiles up to and including the current one and is calculated as (number of events in the quantiles) / (number of events expected by random) - 1. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Note that the value of Gain is the same as the value of Cumulative Lift - 1. If the value of Gain is greater than 0, then your model is better at identifying events than using no model.

Captured Response Percentage



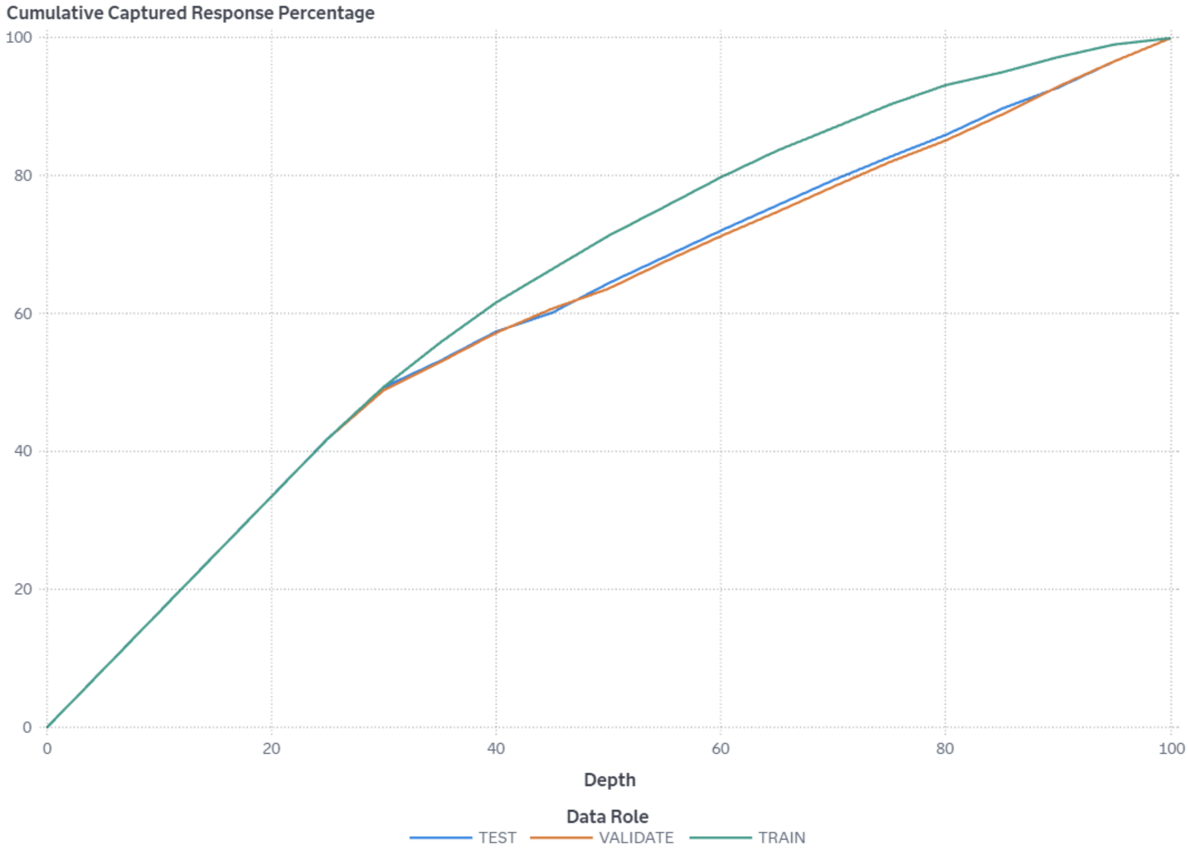
At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TRAIN partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TEST partition has a Captured response percentage of 8.4 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

Captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Captured response percentage is the percentage of the total number of events that are in that quantile. With no model, it is expected that 5% of the events are in each quantile.

Cumulative Captured Response Percentage



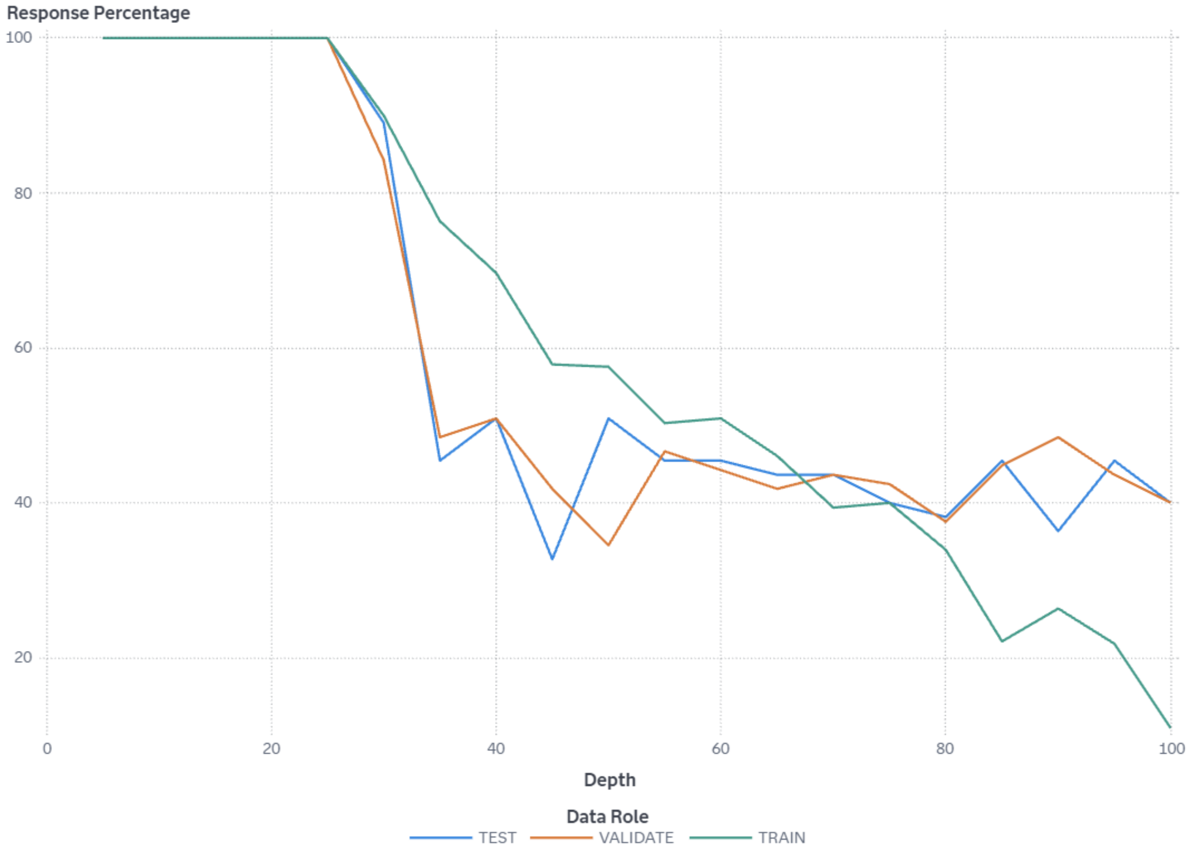
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative captured response percentage of 16.8 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.77.

Cumulative captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative captured response percentage for a particular quantile is the percentage of the total number of events that are in the quantiles up to and including the current quantile. With no model, it is expected that 5% of the events are in each quantile, so the cumulative captured response percentage at depth 10 would be 10%.

Response Percentage



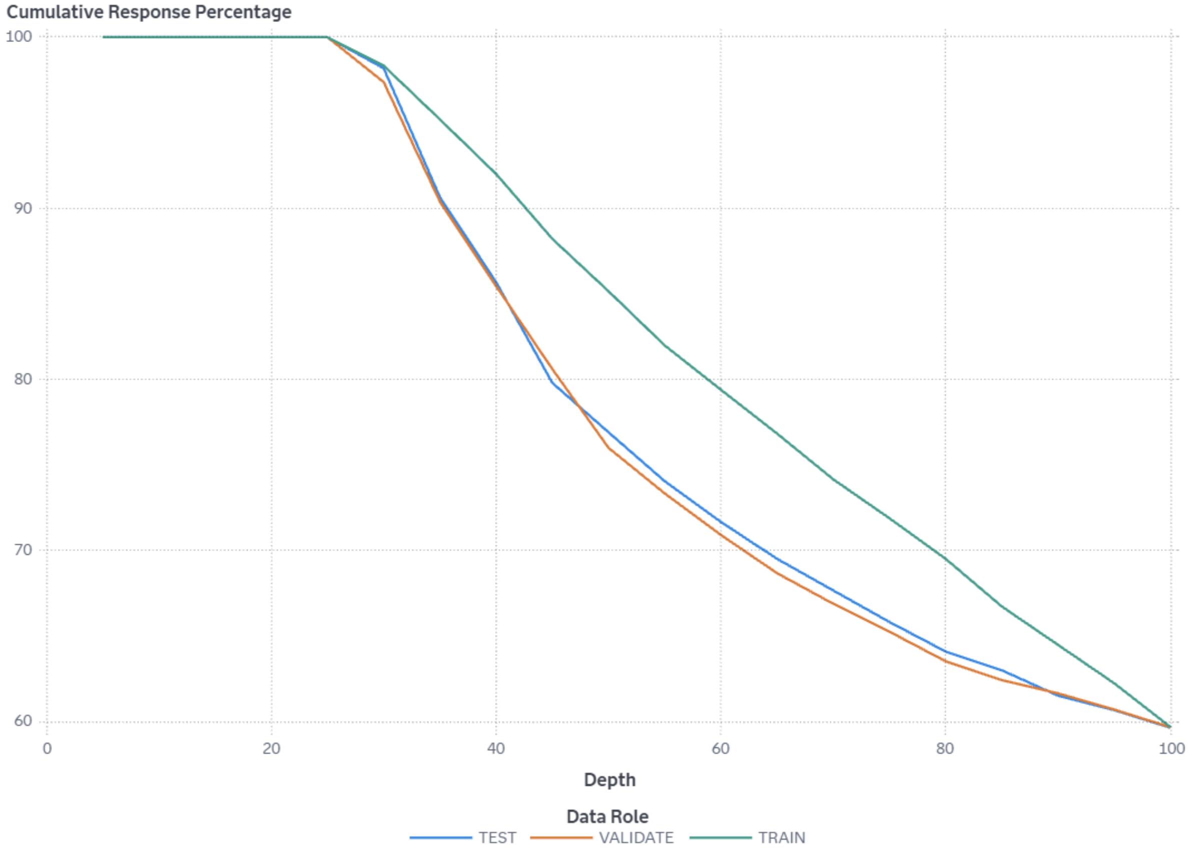
At the 5% quantile (depth of 5), the VALIDATE partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TRAIN partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TEST partition has a Response percentage of 100. The best possible value of Response percentage for this partition at depth 5 is 100.

Response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Response percentage is the percentage of observations that are events in that quantile. With no model, it is expected that the response percentage is constant across quantiles, 100*overall-event-rate. This is also called the baseline response percentage.

Cumulative Response Percentage



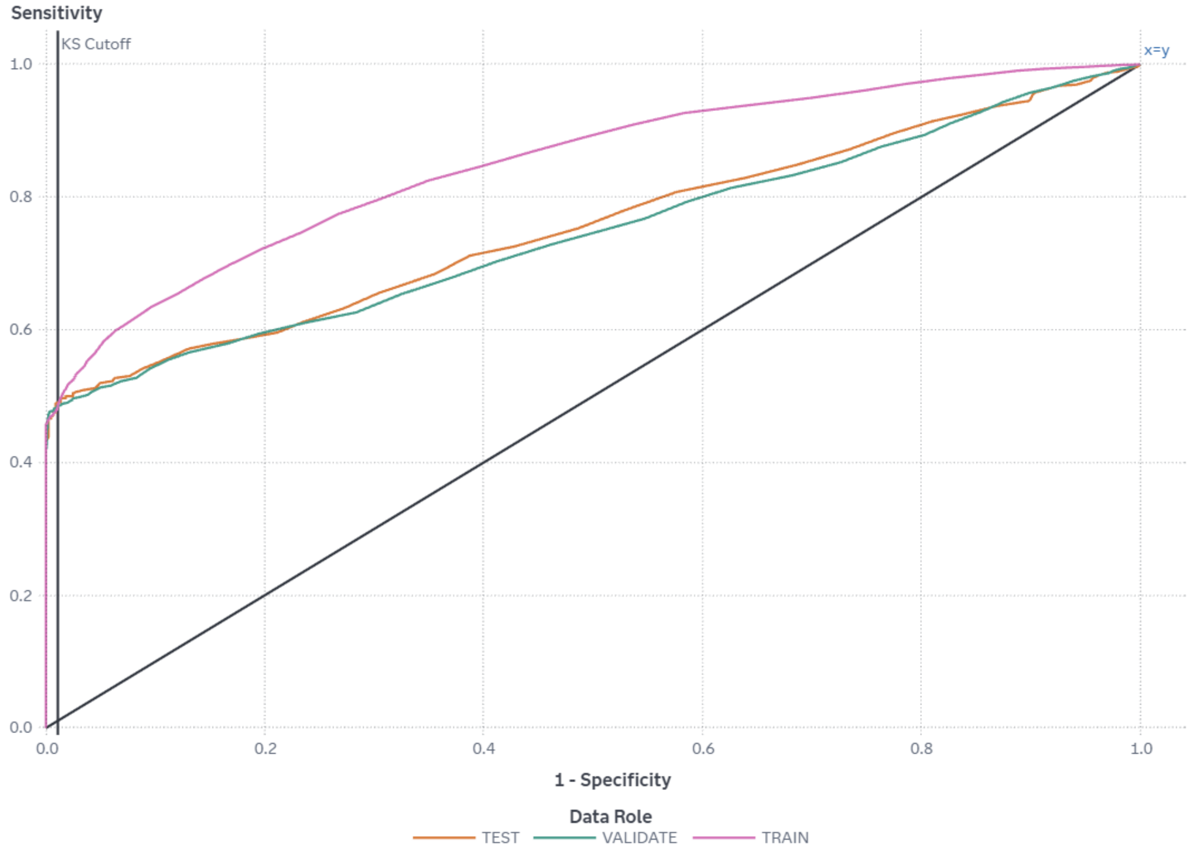
In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative response percentage of 100. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

Cumulative response percentage is calculated by sorting in descending order each partition of the data by the predicted probability of the target event P_Reached_on_Time_Y_N1, which represents the predicted probability of the event "1" for the target Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative response percentage for a particular quantile is the percentage of observations that are events in the quantiles up to and including the current quantile. With no model, it is expected that the response percentage is constant across quantiles, 100*overall-event-rate. This is also called the baseline response percentage.

ROC



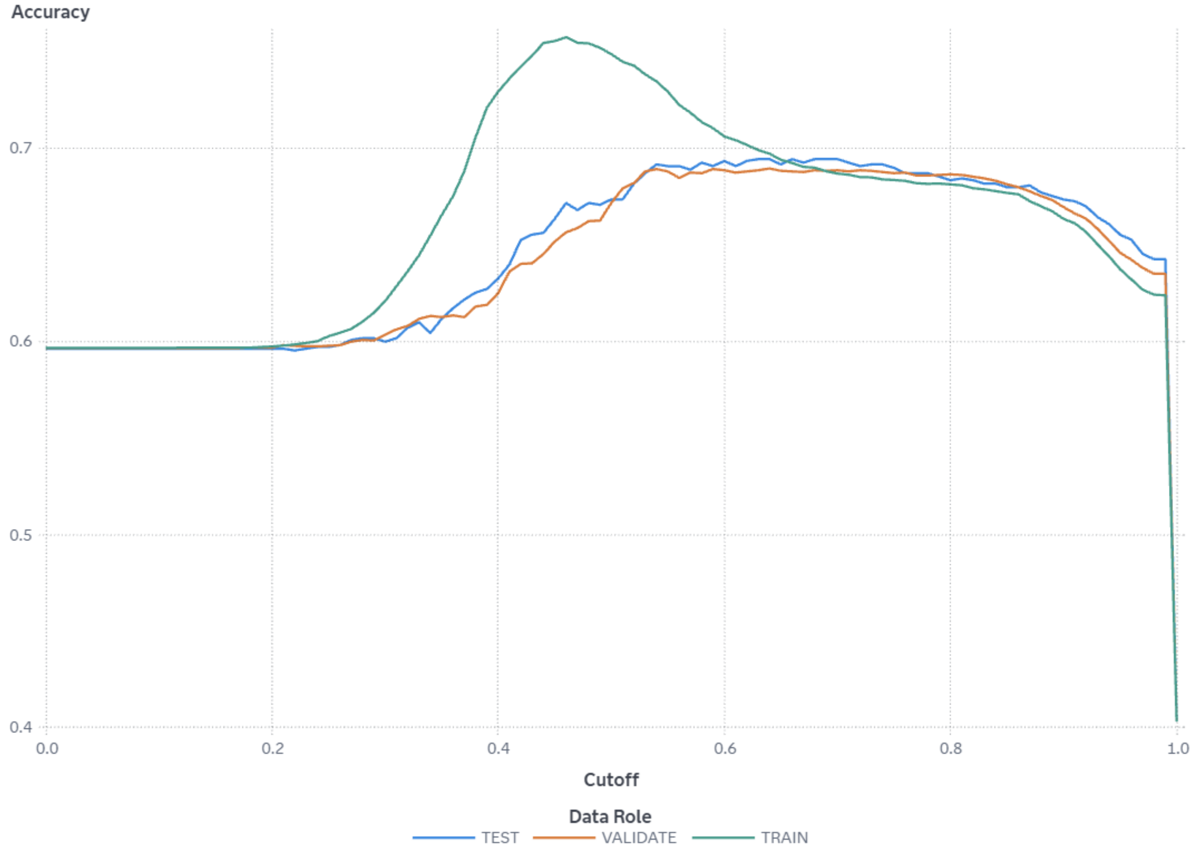
The ROC curve is a plot of sensitivity (the true positive rate) against 1-specificity (the false positive rate), which are both measures of classification based on the confusion matrix. These measures are calculated at various cutoff values. To help identify the best cutoff to use when scoring your data, the KS Cutoff reference line is drawn at the value of 1-specificity where the greatest difference between sensitivity and 1-specificity is observed for the VALIDATE partition. The KS Cutoff line is drawn at the cutoff value 0.72, where the 1-specificity value is 0.011 and the sensitivity value is 0.486.

Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event. Sensitivity is calculated as TP / (TP + FN). Specificity, the true negative rate, is calculated as TN / (TN + FP), so 1-specificity is FP / (TN + FP). The values of sensitivity and 1-specificity are plotted at each cutoff value.

A ROC curve that rapidly approaches the upper-left corner of the graph, where the difference between sensitivity and 1-specificity is the greatest, indicates a more accurate model. A diagonal line where sensitivity = 1-specificity indicates a random model.

Accuracy



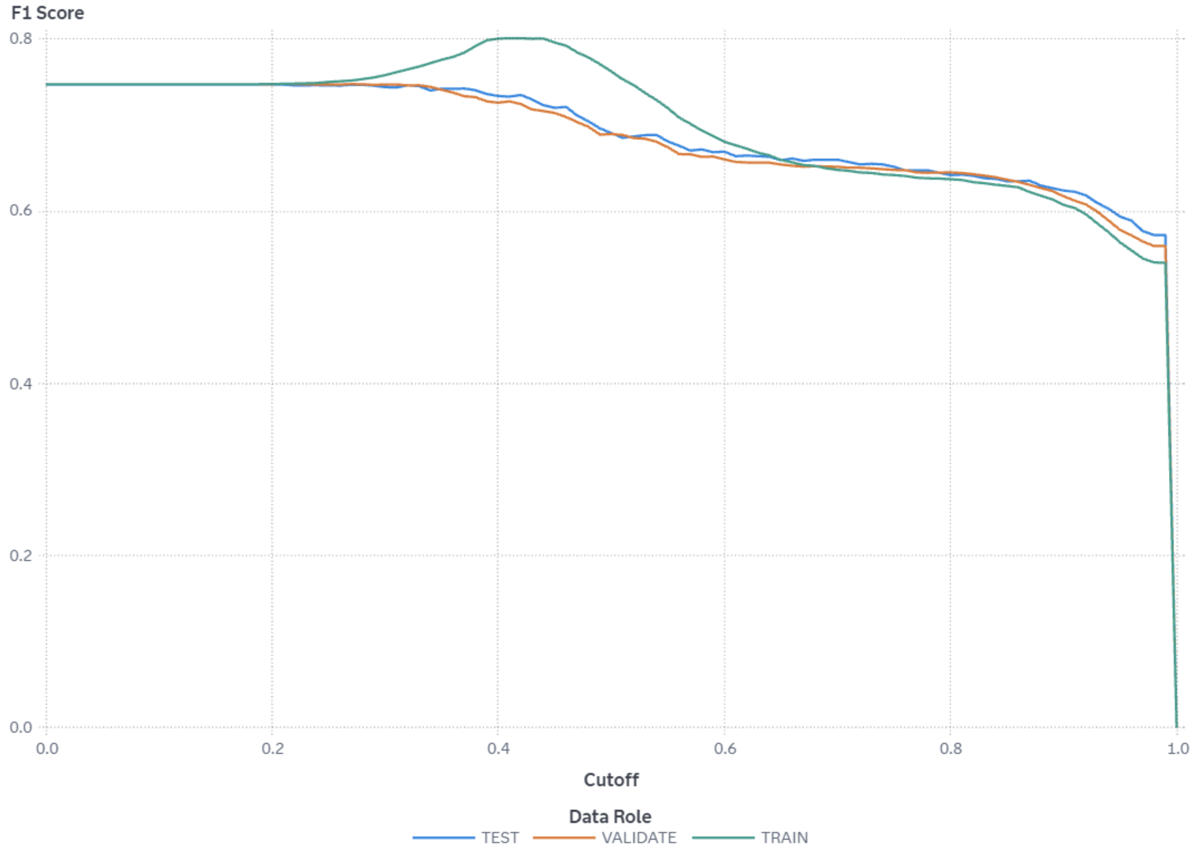
For this model, the accuracy in the TEST partition at the cutoff of 0.5 is 0.674.

For this model, the accuracy in the TRAIN partition at the cutoff of 0.5 is 0.749.

For this model, the accuracy in the VALIDATE partition at the cutoff of 0.5 is 0.672.

Accuracy is the proportion of observations that are correctly classified as either an event or non-event, calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event. When the predicted classification and the actual classification are both events (true positives) or both non-events (true negatives), the observation is correctly classified. If the predicted classification and actual classification disagree, then the observation is incorrectly classified. Accuracy is calculated as (true positives + true negatives) / (total observations).

F1 Score



For this model, the F1 score in the TEST partition at the cutoff of 0.5 is 0.691.

For this model, the F1 score in the TRAIN partition at the cutoff of 0.5 is 0.762.

For this model, the F1 score in the VALIDATE partition at the cutoff of 0.5 is 0.69.

The F1 score combines the measures of precision and recall (or sensitivity), which are measures of classification based on the confusion matrix that are calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether P_Reached_on_Time_Y_N1, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When P_Reached_on_Time_Y_N1 is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event.

Precision is calculated as TP / (TP + FP), and recall (or sensitivity) is calculated as TP / (TP + FN). The F1 score is calculated as 2*Precision*Recall / (Precision + Recall), which is the harmonic mean of Precision and Recall. Larger F1 scores indicate a more accurate model.

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
Reached.on.Time_Y.N	TEST	2	2
Reached.on.Time_Y.N	TRAIN	1	1
Reached.on.Time_Y.N	VALIDATE	0	0

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,100	0.1793	1,100	0.4235
6,599	0.1602	6,599	0.4002
3,300	0.1806	3,300	0.4250

Misclassification Rate	Multi-Class Log Loss	KS (Youden)	Area Under ROC
0.3264	0.5021	0.4834	0.7545
0.2511	0.4632	0.5382	0.8497
0.3282	0.5048	0.4750	0.7452

Gini Coefficient	Gamma	Tau	KS Cutoff
0.5090	0.5178	0.2452	0.6800
0.6994	0.7102	0.3367	0.5200
0.4905	0.4989	0.2361	0.7200

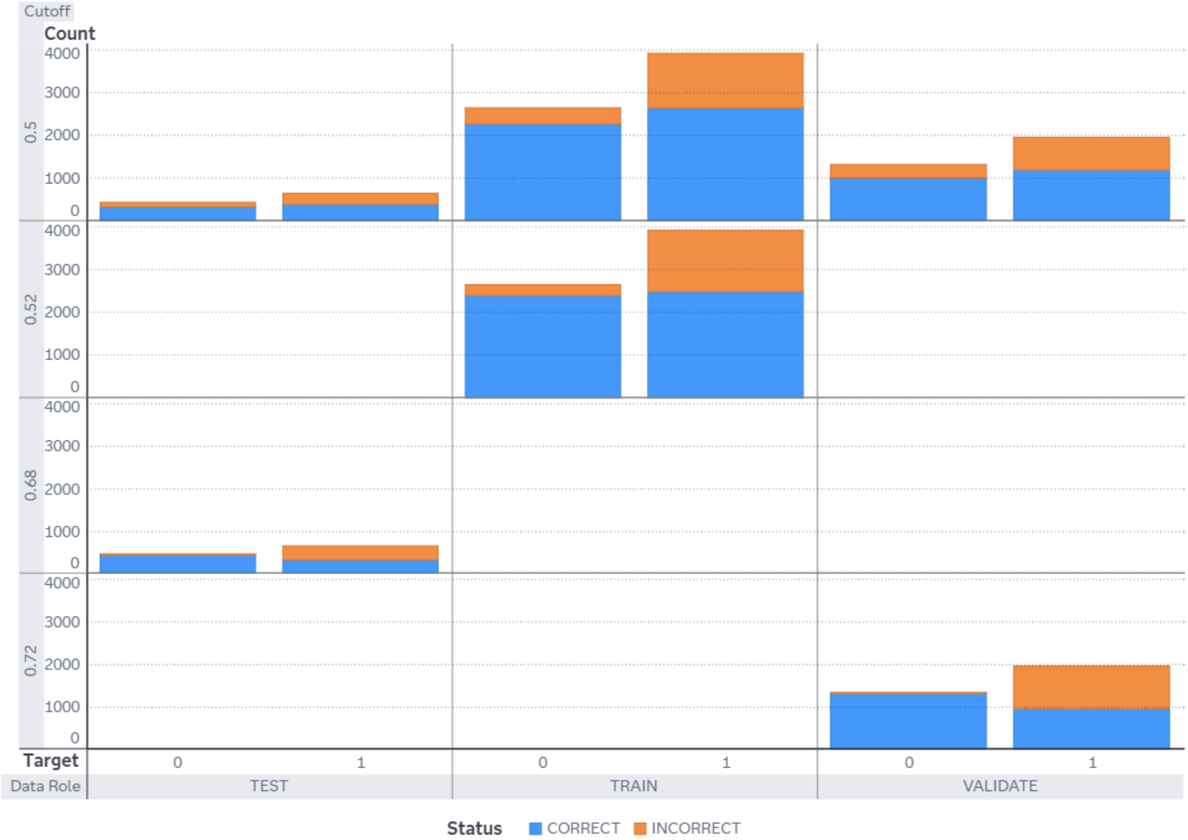
KS at User-Specified Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0.3770	0.3055	0.3264
0.5333	0.2570	0.2511
0.3726	0.3112	0.3282

Percentage Plot

The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.52 (TRAIN), 0.72 (VALIDATE), 0.68 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Count Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0.52 (TRAIN), 0.72 (VALIDATE), 0.68 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Table

Cutoff	Cutoff Source	Target Name	Response
0.5000	Default	Reached.on.Time_Y.N	CORRECT
0.5000	Default	Reached.on.Time_Y.N	INCORRECT
0.5000	Default	Reached.on.Time_Y.N	CORRECT
0.5000	Default	Reached.on.Time_Y.N	INCORRECT
0.5200	KS	Reached.on.Time_Y.N	CORRECT
0.5200	KS	Reached.on.Time_Y.N	INCORRECT
0.5200	KS	Reached.on.Time_Y.N	CORRECT
0.5200	KS	Reached.on.Time_Y.N	INCORRECT
0.6800	KS	Reached.on.Time_Y.N	CORRECT
0.6800	KS	Reached.on.Time_Y.N	INCORRECT
0.6800	KS	Reached.on.Time_Y.N	CORRECT
0.6800	KS	Reached.on.Time_Y.N	INCORRECT
0.7200	KS	Reached.on.Time_Y.N	CORRECT
0.7200	KS	Reached.on.Time_Y.N	INCORRECT
0.7200	KS	Reached.on.Time_Y.N	CORRECT
0.7200	KS	Reached.on.Time_Y.N	INCORRECT

Event	Value	Training Frequency	Validation Frequency

Event	Value	Training Frequency	Validation Frequency
1	True Positive	2,658	1,204
1	False Negative	1,280	765
0	True Negative	2,284	1,013
0	False Positive	377	318
1	True Positive	2,497	
1	False Negative	1,441	
0	True Negative	2,406	
0	False Positive	255	
1	True Positive		
1	False Negative		
0	True Negative		
0	False Positive		
1	True Positive		956
1	False Negative		1,013
0	True Negative		1,317
0	False Positive		14

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
401	67.4962	61.1478	61.1280
255	32.5038	38.8522	38.8720
340	85.8324	76.1082	76.5766
104	14.1676	23.8918	23.4234
	63.4078		
	36.5922		
	90.4171		
	9.5829		
326			49.6951
330			50.3049
438			98.6486
6			1.3514

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
		48.5526	
		51.4474	
		98.9482	
		1.0518	

Properties

Property Name	Property Value
atAppendLookup	false
atCreateHistory	false
atHistoryLibUri	
atHistoryTblName	
atLeaveAutotuneOn	false
atLookupTableUri	
atMaxBayes	100
atMaxEval	50
atMaxIter	5
atMaxTime	60
atObjectiveInt	ASE
atObjectiveNom	KS
atPopSize	10
atSampleSize	50
atSearchMethod	GA
atTrainProp	0.7000
atUpdateProperties	false
atUseLookup	false
atValidFold	5
atValidMethod	PARTITION
atValidProp	0.3000
atbagFreqInitLgbm	0
atbagFreqLBLgbm	0
atbagFreqLgbm	true
atbagFreqUBLgbm	7
atbagPctInitLgbm	0.5000
atbagPctLBLgbm	0.2000
atbagPctLgbm	true
atbagPctUBLgbm	0.9500

Property Name	Property Value
atinputPctInitLgbm	1
atinputPctLBLgbm	0.1000
atinputPctLgbm	true
atinputPctUBLgbm	1
atintervalBins	true
atintervalBinsInit	50
atintervalBinsLB	20
atintervalBinsUB	100
atlasso	true
atlassoInit	0
atlassoLB	0
atlassoUB	10
atleafSize	false
atleafSizeInit	5
atleafSizeLB	1
atleafSizeUB	100
atlearnrt	true
atlearnrtInit	0.1000
atlearnrtLB	0.0100
atlearnrtUB	1
atmaxdepth	true
atmaxdepthInit	4
atmaxdepthLB	1
atmaxdepthUB	6
atntrees	true
atntreesInit	100
atntreesLB	20
atntreesUB	150
atridge	true
atridgeInit	1
atridgeLB	0

Property Name	Property Value
atridgeUB	10
atsamprt	true
atsamprtInit	0.5000
atsamprtLB	0.1000
atsamprtUB	1
atvarsToTry	true
atvarsToTryInit	100
atvarsToTryLB	1
atvarsToTryUB	100
autotune_enabled	false
bagFractionLgbm	0.5000
bagFreqLgbm	0
binaryProbCutoff	0.5000
boostingLgbm	GBDT
classDistrLgbm	MULTICLASS
codeLocation	mlearning
dataMiningVersion	V2025.08
defaultVarsPerTree	true
deterministicLgbm	false
distribution	GAUSSIAN
earlyStop	true
earlyStopMethod	STAGNATION
esMetric	MCR
esMinimum	false
esThreshold	0
esThresholdIter	0
exactPctlLift	true
explainFidelity	false
explainInfo	false
fullDatasetReconstitution	false

Property Name	Property Value
icePlots	false
inputFractionLgbm	1
intBinMethod	QUANTILE
intervalBins	50
intervalDistrLgbm	REGRESSION
lasso	0
learningRate	0.1000
lightGBM_enabled	false
maxBranch	2
maxCategories	128
maxDepth	4
maxNumShapVars	20
minLeafSize	5
minUseInSearch	1
missingLgbm	true
missingValue	USEINSEARCH
nBins	50
ntrees	100
pdNumImportantInputs	5
pdObsSamples	1,000
pdPlots	false
performKernelShap	false
performLime	false
performVI	false
power	1.5000
ridge	1
seed	12,345
seedId	12,345
specifyRows	RANDOM
stagnation	5

Property Name	Property Value
subsampleRate	0.5000
templateRevision	5
tolerance	0
train	true
truncateLl	5
truncateUl	95
userProbCutoff	false
varsToTry	100

Output

The SAS System

The GRADBOOST Procedure

Model Information	
Number of Trees	100
Learning Rate	0.1
Subsampling Rate	0.5
Number of Variables Per Split	10
Number of Bins	50
Number of Input Variables	10
Maximum Number of Tree Nodes	31
Minimum Number of Tree Nodes	15
Maximum Number of Branches	2
Minimum Number of Branches	2
Maximum Depth	4
Minimum Depth	4
Maximum Number of Leaves	16
Minimum Number of Leaves	8
Maximum Leaf Size	2339
Minimum Leaf Size	5
Seed	12345
Lasso (L1) penalty	0
Ridge (L2) penalty	1
Actual Number of Trees	100
Average Number of Leaves	12.41
Early stopping stagnation	5
Early stopping threshold	0
Early stopping threshold iterations	0
Early stopping tolerance	0

	Training	Validation	Test	Total
Number of Observations Read	6599	3300	1100	10999
Number of Observations Used	6599	3300	1100	10999

Variable Importance

Variable	Importance	Std Dev Importance	Relative Importance
Discount_offered	24.7270	54.7061	1.0000
Weight_in_gms	6.7128	7.5593	0.2715
Prior_purchases	5.3399	8.1657	0.2160
Cost_of_the_Product	4.3474	3.0244	0.1758
Customer_rating	2.5987	2.4439	0.1051
Customer_care_calls	2.4959	2.3762	0.1009
Warehouse_block	2.0633	2.0216	0.0834
Product_importance	0.8122	1.3192	0.0328
Mode_of_Shipment	0.6460	1.1762	0.0261
Gender	0.4156	0.9089	0.0168

Fit Statistics

Number of Trees	Training Average Square Error	Validation Average Square Error	Test Average Square Error	Training Misclassification Rate	Validation Misclassification Rate	Test Misclassification Rate	Training Log Loss	Validation Log Loss	Test Log Loss
1	0.229	0.229	0.229	0.403	0.403	0.404	0.650	0.650	0.650
2	0.220	0.219	0.220	0.403	0.403	0.404	0.630	0.629	0.629
3	0.212	0.212	0.212	0.403	0.403	0.404	0.614	0.612	0.613
4	0.206	0.205	0.206	0.403	0.403	0.404	0.599	0.598	0.599
5	0.202	0.201	0.201	0.403	0.403	0.404	0.588	0.586	0.587
6	0.197	0.196	0.197	0.385	0.396	0.382	0.577	0.575	0.577
7	0.194	0.193	0.194	0.365	0.386	0.385	0.568	0.566	0.567
8	0.191	0.190	0.191	0.346	0.358	0.377	0.560	0.558	0.559
9	0.189	0.188	0.188	0.332	0.348	0.366	0.553	0.550	0.552
10	0.187	0.186	0.187	0.324	0.342	0.357	0.547	0.544	0.546
11	0.185	0.184	0.185	0.321	0.336	0.362	0.542	0.539	0.541
12	0.184	0.183	0.184	0.323	0.331	0.353	0.538	0.535	0.537
13	0.183	0.182	0.183	0.315	0.326	0.341	0.534	0.531	0.533
14	0.182	0.181	0.182	0.307	0.321	0.327	0.531	0.528	0.530
15	0.181	0.180	0.181	0.306	0.319	0.326	0.527	0.524	0.526
16	0.181	0.180	0.181	0.308	0.322	0.323	0.525	0.522	0.523
17	0.180	0.180	0.180	0.308	0.320	0.319	0.522	0.520	0.521
18	0.180	0.179	0.180	0.308	0.319	0.320	0.520	0.518	0.519
19	0.179	0.179	0.180	0.307	0.315	0.318	0.518	0.516	0.517
20	0.179	0.179	0.180	0.306	0.315	0.316	0.516	0.514	0.516
21	0.178	0.178	0.179	0.306	0.313	0.319	0.514	0.513	0.514
22	0.178	0.178	0.179	0.303	0.317	0.320	0.513	0.511	0.513
23	0.178	0.178	0.179	0.302	0.316	0.317	0.511	0.510	0.512
24	0.178	0.178	0.179	0.304	0.318	0.315	0.510	0.509	0.511
25	0.177	0.178	0.179	0.306	0.316	0.320	0.508	0.508	0.510
26	0.177	0.178	0.179	0.303	0.315	0.314	0.507	0.507	0.509
27	0.177	0.178	0.179	0.302	0.313	0.315	0.506	0.507	0.509
28	0.177	0.178	0.179	0.301	0.312	0.318	0.505	0.506	0.508
29	0.176	0.178	0.179	0.301	0.314	0.315	0.504	0.505	0.508
30	0.176	0.177	0.179	0.300	0.311	0.315	0.503	0.504	0.507
31	0.176	0.177	0.179	0.300	0.312	0.311	0.502	0.503	0.506
32	0.176	0.177	0.179	0.302	0.312	0.311	0.502	0.503	0.506
33	0.176	0.177	0.179	0.302	0.312	0.312	0.501	0.502	0.506
34	0.175	0.177	0.179	0.302	0.311	0.317	0.500	0.502	0.505
35	0.175	0.177	0.179	0.302	0.311	0.317	0.500	0.502	0.505
36	0.175	0.177	0.179	0.302	0.310	0.321	0.499	0.502	0.505
37	0.175	0.178	0.179	0.300	0.309	0.318	0.498	0.502	0.504
38	0.175	0.177	0.179	0.299	0.312	0.323	0.497	0.501	0.504
39	0.174	0.177	0.179	0.300	0.311	0.318	0.496	0.500	0.503
40	0.174	0.177	0.179	0.299	0.312	0.317	0.495	0.500	0.503
41	0.174	0.177	0.179	0.297	0.314	0.319	0.495	0.500	0.503
42	0.174	0.177	0.178	0.295	0.314	0.318	0.494	0.500	0.502
43	0.173	0.177	0.178	0.295	0.314	0.315	0.493	0.500	0.502
44	0.173	0.177	0.178	0.293	0.313	0.315	0.493	0.500	0.502
45	0.173	0.177	0.178	0.292	0.313	0.322	0.493	0.500	0.502
46	0.173	0.178	0.179	0.293	0.313	0.317	0.492	0.500	0.502
47	0.173	0.177	0.178	0.291	0.310	0.318	0.492	0.500	0.502
48	0.173	0.177	0.178	0.293	0.313	0.317	0.491	0.500	0.501
49	0.172	0.178	0.179	0.289	0.315	0.321	0.490	0.500	0.502
50	0.172	0.178	0.179	0.289	0.315	0.316	0.490	0.500	0.502
51	0.172	0.178	0.179	0.288	0.322	0.323	0.489	0.500	0.502
52	0.172	0.178	0.179	0.289	0.319	0.321	0.489	0.500	0.502
53	0.171	0.178	0.179	0.288	0.317	0.326	0.488	0.500	0.502
54	0.171	0.178	0.179	0.284	0.318	0.324	0.487	0.500	0.501
55	0.171	0.178	0.179	0.284	0.317	0.320	0.487	0.500	0.501
56	0.170	0.178	0.179	0.282	0.318	0.322	0.486	0.501	0.501
57	0.170	0.179	0.179	0.280	0.320	0.323	0.485	0.501	0.501
58	0.170	0.179	0.179	0.278	0.319	0.324	0.484	0.501	0.501
59	0.169	0.179	0.179	0.279	0.318	0.323	0.484	0.501	0.501
60	0.169	0.179	0.179	0.278	0.317	0.319	0.483	0.501	0.501
61	0.169	0.179	0.179	0.277	0.317	0.322	0.483	0.501	0.501
62	0.169	0.179	0.179	0.277	0.318	0.317	0.482	0.501	0.501
63	0.169	0.179	0.179	0.276	0.318	0.321	0.482	0.501	0.501
64	0.168	0.179	0.179	0.276	0.317	0.321	0.482	0.501	0.501
65	0.168	0.179	0.179	0.275	0.316	0.322	0.481	0.501	0.501
66	0.168	0.179	0.179	0.275	0.319	0.320	0.480	0.501	0.501
67	0.168	0.179	0.179	0.272	0.318	0.322	0.480	0.501	0.501
68	0.167	0.179	0.178	0.273	0.319	0.322	0.479	0.501	0.500
69	0.167	0.179	0.178	0.273	0.319	0.324	0.479	0.501	0.500
70	0.167	0.179	0.178	0.270	0.317	0.327	0.478	0.501	0.500
71	0.166	0.179	0.178	0.267	0.318	0.321	0.477	0.501	0.500
72	0.166	0.179	0.178	0.268	0.320	0.316	0.477	0.501	0.500
73	0.166	0.179	0.178	0.268	0.321	0.322	0.476	0.502	0.500
74	0.166	0.179	0.178	0.266	0.318	0.318	0.476	0.502	0.499
75	0.166	0.179	0.178	0.265	0.317	0.315	0.475	0.501	0.499
76	0.166	0.179	0.178	0.264	0.317	0.324	0.475	0.501	0.500
77	0.165	0.179	0.178	0.265	0.317	0.321	0.475	0.501	0.500
78	0.165	0.179	0.178	0.263	0.321	0.320	0.474	0.502	0.500
79	0.165	0.179	0.178	0.263	0.319	0.315	0.474	0.502	0.500
80	0.165	0.179	0.178	0.264	0.321	0.314	0.473	0.502	0.500
81	0.165	0.179	0.178	0.263	0.322	0.313	0.473	0.503	0.500
82	0.164	0.179	0.178	0.262	0.319	0.322	0.472	0.502	0.500
83	0.164	0.179	0.179	0.260	0.318	0.325	0.472	0.502	0.500
84	0.164	0.179	0.179	0.262	0.319	0.321	0.471	0.502	0.501
85	0.164	0.179	0.179	0.262	0.320	0.323	0.471	0.502	0.501
86	0.164	0.180	0.179	0.262	0.322	0.322	0.471	0.503	0.501
87	0.163	0.180	0.179	0.262	0.321	0.319	0.470	0.503	0.501
88	0.163	0.180	0.179	0.261	0.320	0.326	0.470	0.503	0.501
89	0.163	0.180	0.179	0.259	0.324	0.326	0.469	0.503	0.501
90	0.163	0.180	0.179	0.261	0.325	0.325	0.469	0.503	0.502
91	0.162	0.180	0.179	0.260	0.325	0.324	0.468	0.504	0.502
92	0.162	0.180	0.179	0.260	0.324	0.326	0.468	0.504	0.502
93	0.162	0.180	0.179	0.256	0.327	0.330	0.467	0.505	0.502
94	0.162	0.181	0.179	0.256	0.329	0.326	0.466	0.505	0.502
95	0.161	0.180	0.180	0.256	0.328	0.330	0.466	0.504	0.503
96	0.161	0.181	0.179	0.252	0.327	0.325	0.465	0.505	0.502
97	0.161	0.181	0.179	0.251	0.323	0.329	0.465	0.505	0.502
98	0.161	0.181	0.179	0.253	0.329	0.330	0.464	0.505	0.502
99	0.160	0.181	0.179	0.252	0.328	0.327	0.464	0.505	0.502
100	0.160	0.181	0.179	0.251	0.328	0.326	0.463	0.505	0.502

Output CAS Tables

CAS Library	Name	Number of Rows	Number of Columns
Analytics_Project_863d0c53f2ae-4559-81c5-6981213c1802	_F3Y7FL56OPUG7Y4YRDJRXS8DV_AST	1	2

Predicted Probability Variables

Reached.on.Time_Y.N	Variable
1	P_Reached_on_Time_Y_N1
0	P_Reached_on_Time_Y_N0

Predicted Target Variable

Level Index	Variable
	I_Reached_on_Time_Y_N